

Report on the Meteorological Programme Carried Out During Eleventh Indian Expedition to Antarctica

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Abstract

India Meteorological Department has an on-going meteorological programme in the Antarctic Expeditions. During the eleventh expedition, two new sub programmes viz., Slow rising Low Level Radiosonde and Pilot Balloon ascents were added on experimental basis. In this report data pertaining to surface weather, vertical distribution of ozone and global solar radiation have been presented and discussed. Data on radiometersonde and low level radiosonde ascents and sunphotometer observations could not be included as these are still being processed. Pilot Balloon ascents were not successful due to early bursting of balloon and difficulty in tracking in cold environment.

Introduction

The meteorological programme of India Meteorological Department, in Antarctica, is of on-going nature. During the eleventh expedition, however, two new sub-programmes were included. These are Low Level Slow Rising Radiosonde ascents (LLSR-RS) for determining the temperature and humidity profiles in the lowest few kilometres of the atmosphere and Pilot Balloon (PB) ascents for determining the upper level winds. Both the programmes had been included on an experimental basis.

The on-going observational programmes included continuous surface weather watch, weekly measurement of vertical distribution of ozone, and fortnightly measurement of infra-red radiation fluxes in the upper atmosphere. During onward and return cruise of the expedition from India to Antarctica and back, daily recording of surface and upper air observations formed part of the programme.

Results and Discussions

Presented here are the data pertaining to surface weather, vertical distribution of ozone and global solar radiation. Data on radiometersonde, low level radiosonde ascents are still under process. Pilot balloon ascents were not successful due to early bursting of balloons. Also, tracking of balloon was very difficult due to extreme cold.

Data of surface weather originates from 3 hourly manual synoptic observations and continuous recording of some parameters like pressure, temperature and wind by autographic instruments. Weather charts and satellite cloud Imageries were also received on most of the days.

The climatic data of 1992 are presented in Table I and mean monthly variation in pressure, temperature and wind are shown in Fig 1. Individual meteorological elements are discussed below.

a) Mean sea-level pressure

Mean sea-level pressure varied between 978.7 hPa and 994.9 hPa. Seasonal variation in pressure was not very clear but the trend of alternately rising and falling with humps in summer and winter and sinks in autumn and spring, is discernible. There seems to be some time lag between the seasons and pressure response. During 1992 the lowest pressure of 952.2 hPa was recorded on April 16 and highest pressure of 1015.9 hPa was recorded on June 27.

b) Air temperature

Mean monthly air temperature shows systematic variation with highest mean value of +1.3 °C in January and lowest mean value of -18.9°C in August. It decreased continuously from January to August and increased rapidly thereafter.

Average maximum and minimum temperatures also show similar trend. Highest maximum temperature of +7.8°C was recorded on January 2 and lowest minimum temperature of -31.7°C was recorded on August 29.

Day today temperature variation at Maitri is governed by two factors viz., (i) day and night cycle and (ii) warm air advection from northern latitudes in the wake of depressions. In exclusion of the second factor, diurnal variation (Fig 2) is small but rhythmic in equinoxial months (4-5°C). Minimum temperature is attained by about 0600 UTC (0640 LAT) and maximum is attained by 1200 UTC. During summer solstice the diurnal variation is comparatively more pronounced (7 °C). In peak summer month when the Sun shines continuously, minimum temperature is attained between 0300 and 0600 UTC, and maximum

Table 1: Climatic data of Maitri, Antarctica (1992); Lat 70 ° 45'S,11'44'E																	
Month	Average Mean Sea Level Pressure (hPa)	Temperature °C						Wind (knots)				Mean daily cloudi- ness (Octas)	Number of day with			Num- ber of Bliz- zards	Duration of longest Blizzard (hours)
		Daily mean	Mean daily max.	Mean daily min.	Daily range	Extreme Max. Min.		Pre- vailing direc- tion	Mean daily speed	Abso- lute max.	Number of days with speed ≥ 23		Snow- fall	Fog	Bliz- zard		
Jan	989.4	+ 1.3	+ 4.1	-2.1	6.2	+7.8	-6.4	ESE	14	70	3	5.4	6	-	-	-	-
Feb	991.1	-1.8	+ 0.7	-4.7	5.4	+5.3	-8.4	SE	14	68	3	4.5	4	-	-	-	-
Mar	984.3	-5.9	-3.8	-8.5	4.7	+1.0	-12.2	SE	20	78	12	5.7	8	-	13	5	79
Apr	984.6	-10.7	-8.4	-13.9	5.5	-2.2	-22.3	SE	17	62	8	3.9	8	-	12	4	89
May	990.3	- 10.9	-8.4	-13.9	5.5	-1.5	-22.5	SE	21	71	16	4.2	2	-	6	2	81
Jun	994.9	-12.7	-9.7	-16.4	6.7	-4.0	-25.2	SE	18	73	8	3.5	3	-	1	1	103
Jul	982.3	-16.0	-13.1	-18.8	5.7	-2.0	-28.0	SE	15	53	6	4.1	13	-	6	3	90
Aug	978.7	-18.9	-16.0	-22.8	6.8	-6.8	-31.7	SE	14	80	4	3.4	8	-	6	2	59
Sep	985.0	-18.3	-14.9	-22.1	7.2	-9.0	-27.7	SE	13	60	4	3.2	1	-	5	2	67
Oct	986.5	-12.8	-9.5	-16.4	6.9	-3.0	-24.5	SE	15	64	6	3.7	5	-	8	4	44
Nov	979.5	-6.0	-3.0	-10.2	7.2	+3.2	-15.0	ESE	13	69	2	3.7	7	-	3	1	42
Dec	984.2	-0.8	-1.5	-3.7	2.2	+5.0	-7.7	ESE	13	56	4	5.2	6	2	2	1	24
Annual Total	-		-	-	-	-	-	-	-	-	76	-	71	2	62	25	-
Annual Mean	985.9	-9.4	-	-	5.8	-	-	-	16	-	6.7	4.2	6	-	5	2	-
Range	16.2	17.0	20.1	20.7	5.0	-	-	-	-	-	-	-	-	-	-	-	-

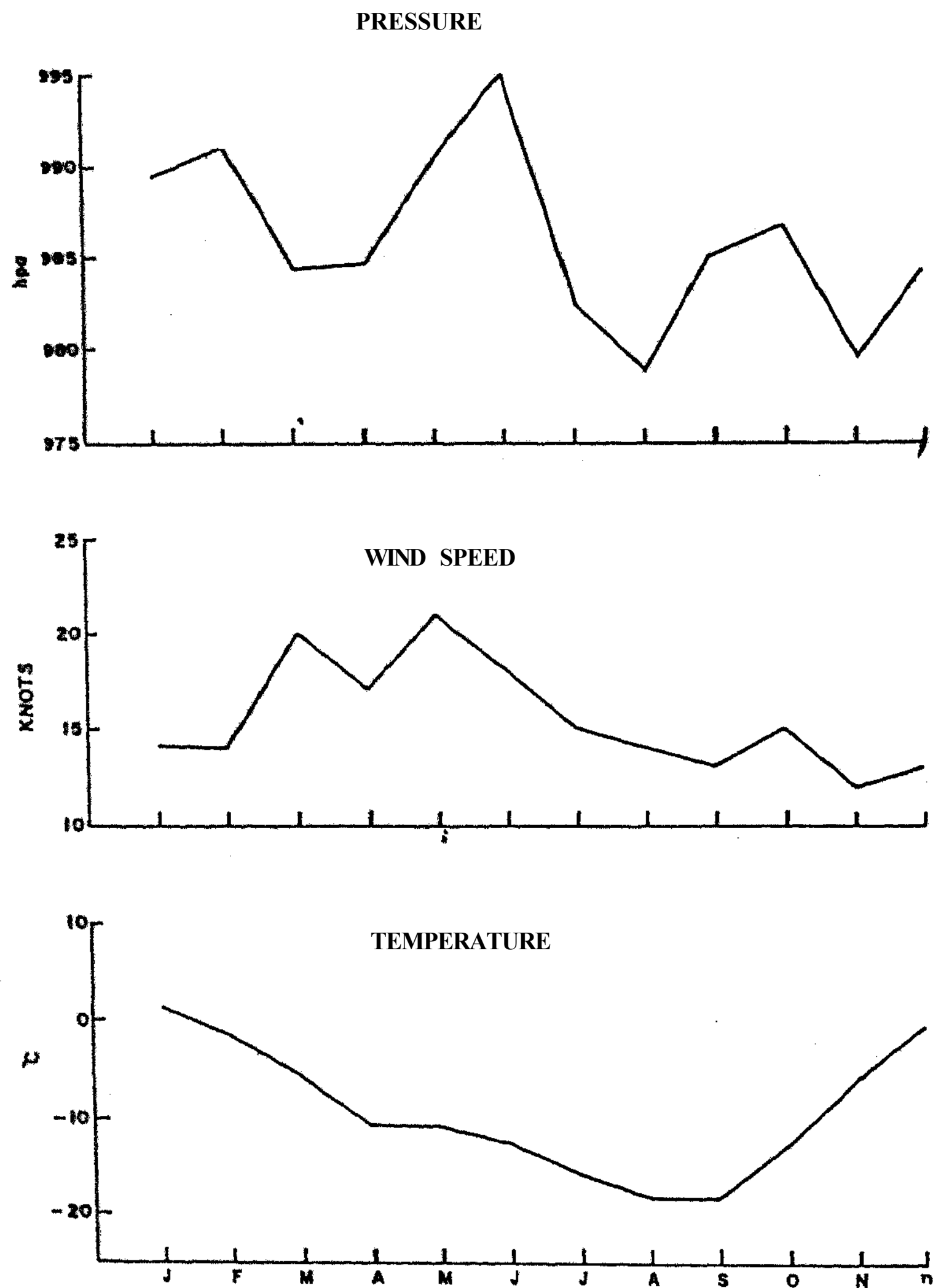
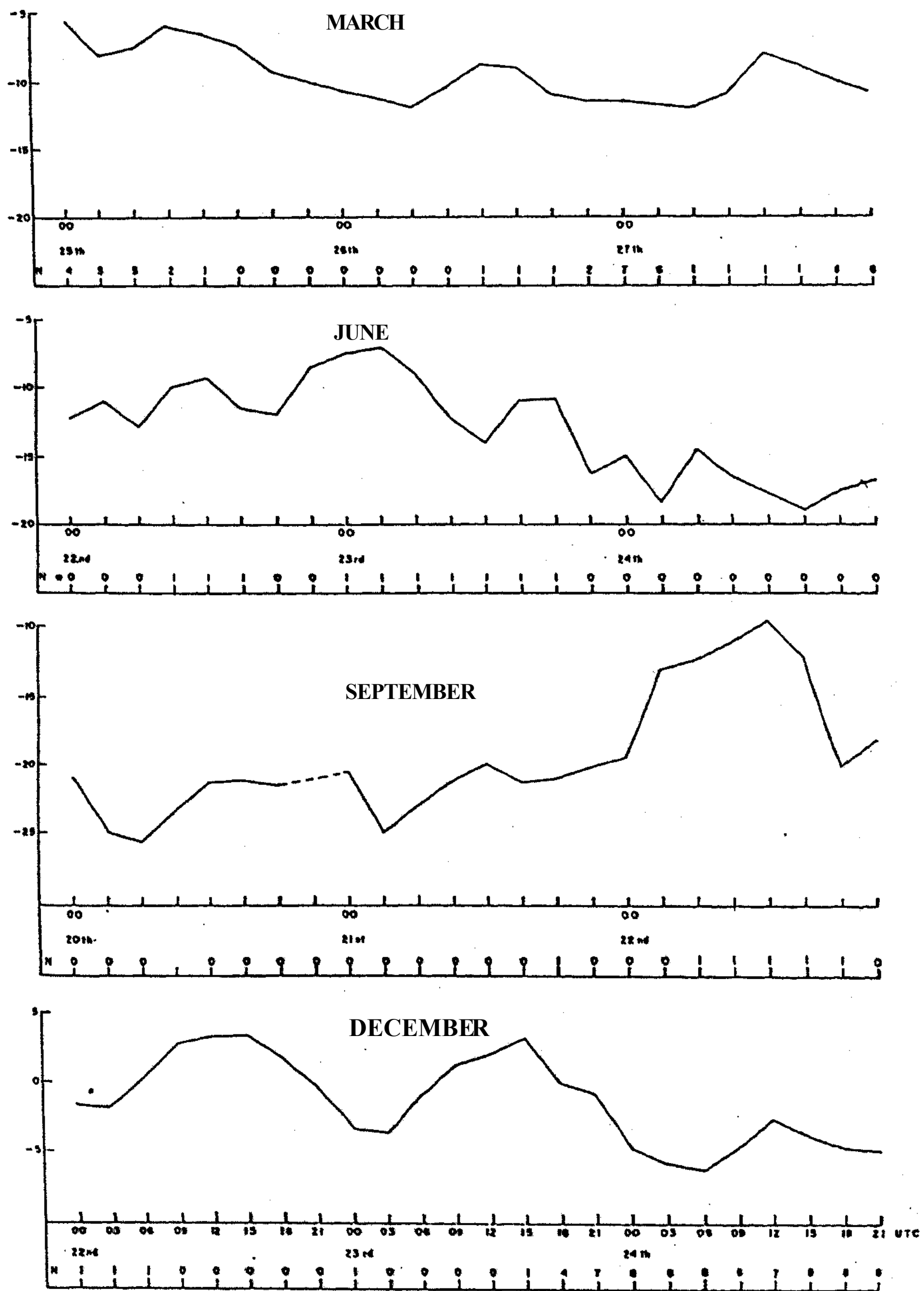


fig 1 : Mean monthly variation of pressure, wind speed and temperature at Maitri during 1992.



N = CLOUD COVER IN OCTAS, TEMP. SCALE ON ORDINATE IN DEGREE CELCIUS.

Fig 2 : Diurnal variation of temperature around equinox and solstice dates as recorded at Maitri in 1992.

temperature is attained between 1200" and 1500 UTC. During June, when the Sun does not appear above horizon, temperature varies at random. Few. hours of twilight everyday between 0800 and 1400 hrs does not cause rhythmic variation in temperature.

Advection of warm air from oceanic areas with approaching depressions, vitiates the diurnal variation of temperature. Maximum and minimum temperatures are attained any time of the day. For this reason they are recorded and set every twelve hours (00 and 12 UT) in the polar region as a standard practice:

c) Surface Wind

Mean monthly wind speed varied between 21 and 12 knots. May was the windiest month with mean daily wind speed of 21 knots and 16 days with wind speed ≥ 23 knots. Highest wind speed recorded during 1992 was 80 knots on August 11.

At Maitri, variation in wind speed follows a characteristic pattern as with most coastal stations of Antarctica. Short periods (1-2 days) of calm or light wind alternate with periods of high wind corresponding to successive passage of low pressure systems.

The prevailing direction is SE closely followed by ESE. Frequency analysis of surface wind directions is given in Table II and depicted in the form of a modified wind rose in Fig 3. On 80 per cent of the occasions wind blows from the SE quadrant. Calm and variable direction account for 8 and 1 per cent, respectively. All other directions put together account for only 11 per cent. Possible explanation for the distinctive groups of directions is given along side the wind rose.

Katabatic winds are an important feature of surface wind in Antarctica. They are characterised by high directional constancy but show large variations in speed. Since these winds blow down an incline under the influence of gravity, their direction at Maitri is invariably southerly as the elevation increases towards south.

An attempt has been made to study the southerly wind at Maitri. Table III, shows the frequency of southerly (dd=16,18 and 20) wind split in two time intervals of the day. Out of the 28.3 per cent occasions of southerly wind, 17.1 per cent were between 2100 and 0600 UTC (night time) while 11.2 per cent: were between 0600 and 2100 UTC (day time). Seasonwise most of the southerly Winds during summer (Nov-Dec-Jan) were between 2100 and 0600 UTC. But in winter the frequency of southerly wind did not show significant variation between the two time intervals.

Table II : Frequency of surface Wind Direction (%) at Maitri, Antarctica (1992) (Total number of observation 2928)																			
Month	CALM 00	NNE 02	NE 05	ENE 07	E 09	ESE 11	SE 14	SSE 16	S 18	SSW 20	SW 23	WSW 25	W 27	WNW 29	NW 32	NNW 34	N 36	VRB 99	Total
Jan	3.6	1.2	0.4	3.2	9.7	35.1	19.4	7.7	10.9	2.4	0.8	-	0.4	-	1.6	-	3.6	-	100.0
Feb	6.9	-	0.4	0.9	5.6	25.0	29.7	8.2	15.5	-	0.4	0.4	2.2	1.3	0.9	0.4	1.7	0.4	99.9
Mar	3.6	-	0.4	1.6	2.0	19.8	31.9	16.9	19.0	0.8	0.4	0.8	1.2	-	0.4	-	1.2	-	100.0
Apr	7.1	-	-	0.4	1.7	14.2	33.3	25.4	14.2	0.8	0.4	-	0.8	-	-	-	0.4	1.3	100.0
May	8.1	-	-	-	1.6	14.9	44.4	11.7	16.9	1.6	0.4	-	-	-	-	-	-	0.4	100.0
Jun	7.5	-	-	-	-	17.1	29.2	15.4	18.8	8.8	1.3	-	0.4	-	0.4	-	-	1.3	100.2
Jul	12.5	-	-	-	1.2	26.6	28.2	13.7	12.9	1.6	-	-	1.6	-	-	-	0.4	1.2	99.9
Aug	14.1	-	-	0.4	1.2	22.6	24.6	12.5	15.7	8.2	0.8	0.8	0.8	0.4	-	-	-	2.8	99.9
Sep	12.5	-	0.8	-	3.8	18.8	29.6	12.9	10.0	2.9	0.8	0.8	3.8	1.7	-	0.4	0.4	0.8	100.0
Oct	9.7	0.4	1.2	0.4	7.3	19.8	26.2	6.5	16.9	2.4	0.8	0.4	2.0	0.8	2.0	-	1.6	1.6	100.0
Nov	8.3	0.8	0.4	1.7	8.8	26.3	24.2	6.3	10.8	2.9	0.4	0.8	1.3	0.8	2.9	0.4	1.7	1.3	100.1
Dec	4.4	1.6	1.6	2.0	13.7	29.4	21.8	5.6	7.7	2.0	0.4	0.4	0.4	-	1.6	0.8	6.1	0.4	99.9
Annual	7.41	0.41	0.38	1.47	5.77	23.02	26.84	12.77	12.88	2.32	0.55	0.51	1.13	0.51	0.65	0.31	2.12	0.96	100.0

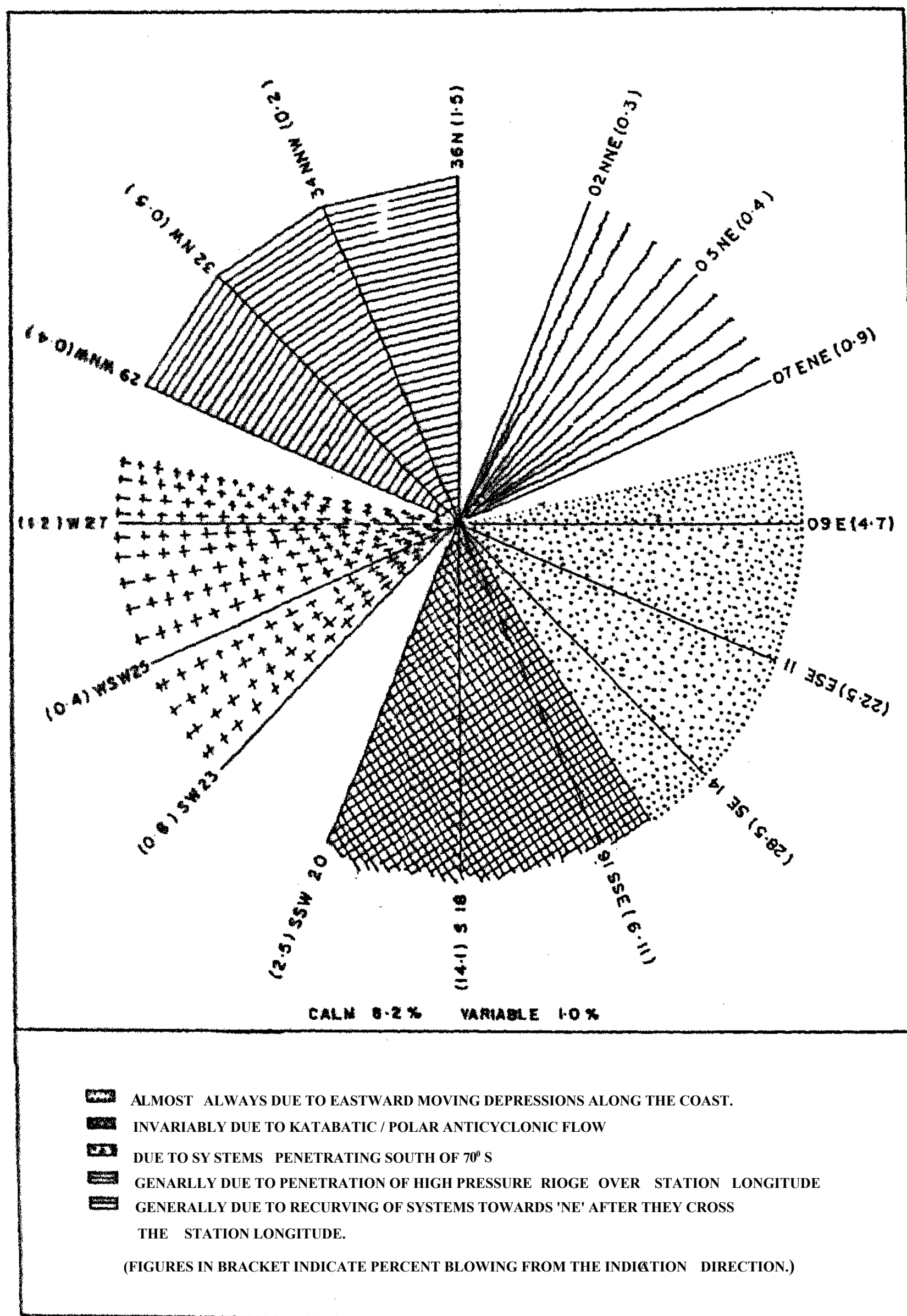


Fig : 1992 Wind Rose data for Maitri station at Antarctica.

Table III: Frequency Analysis of Southerly (dd = 16,18,20) Wind Direction of Maitri**(Figures in the table indicate number of occasions of southerly wind split in two time periods of the day)**

Month	Time period comprising		Total
	21, 00, 03, 06 UTC observations	09, 12, 15, 18 UTC observations	
Jan	46	07	53
Feb	42	13	55
Mar	54	37	91
Apr	46	51	97
May	40	36	76
Jun	44	58	102
Jul	41	29	70
Aug	42	33	75
Sep	32	29	61
Oct	43	19	62
Nov	39	09	48
Dec	32	06	38
Total	501	327	828
%	60.5	39.5	100

Percentage of southerly wind direction = 28.3. (Ref: Wind Rose)

The speed of southerly wind on most occasions varied from 5 to 30 knots. The speed frequently showed large variations over short time intervals (Figs 4a, 4b, 5a, 5b).

d) Blizzards

During 1992 a total of 25 blizzards of varying intensity were experienced at Maitri. While discussing blizzards, it is usually necessary to define the criteria adopted for calling it so as specifying the times of commencement and cessation

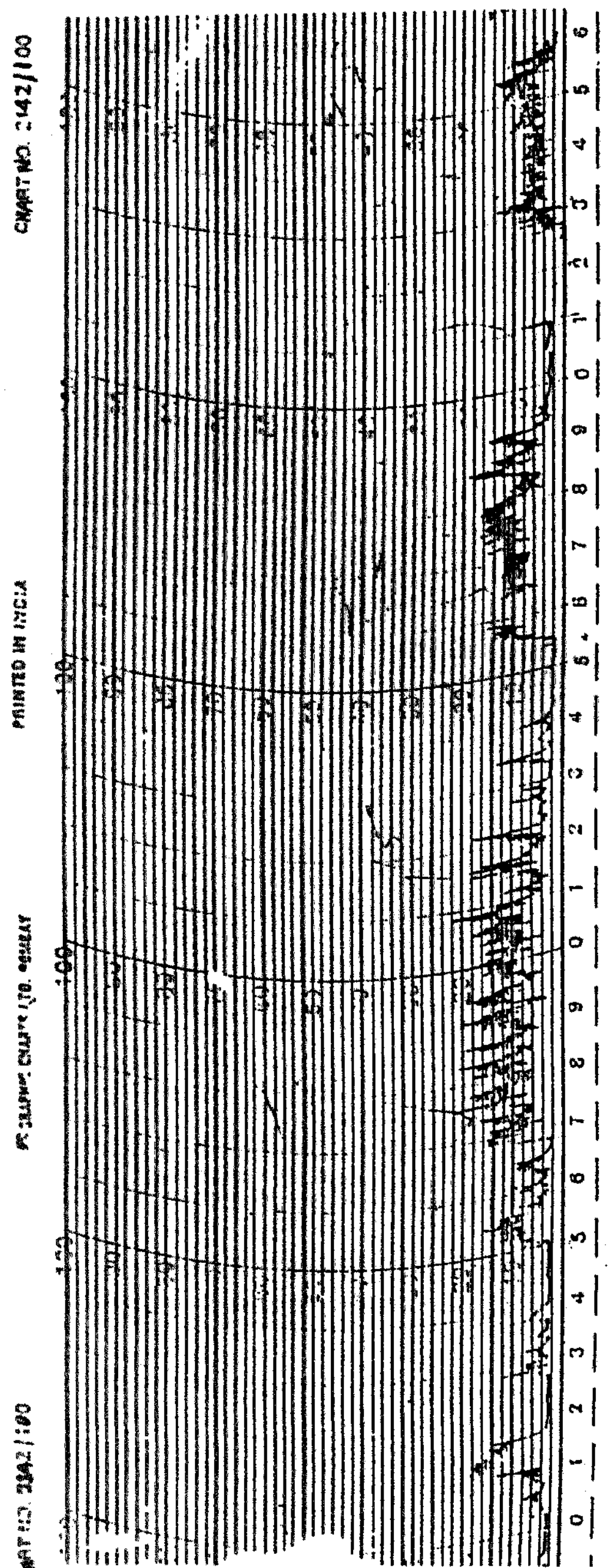


Fig 4a: Typical trace of speed showing large variation over short time interval (21-22 June 1992).

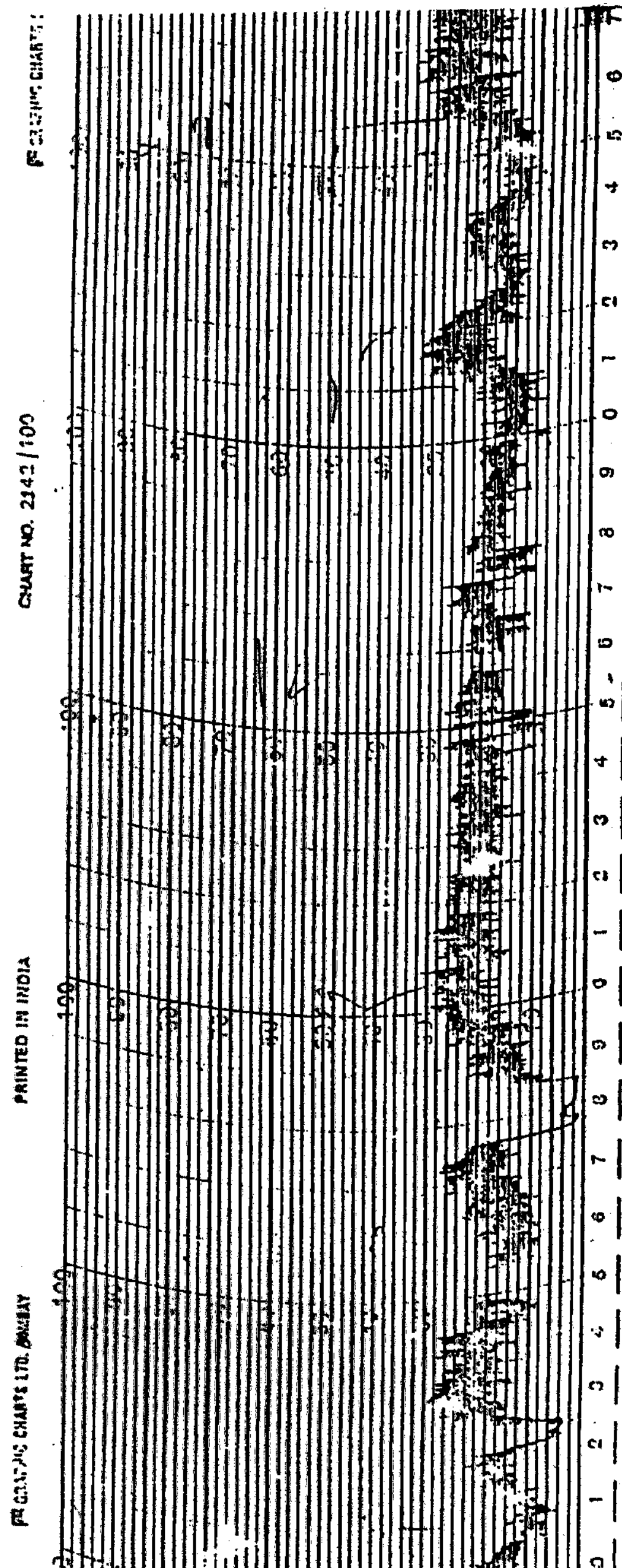


Fig.4b : Typical trace of wind speed showing large variation over short time interval (21-22 June 1992).

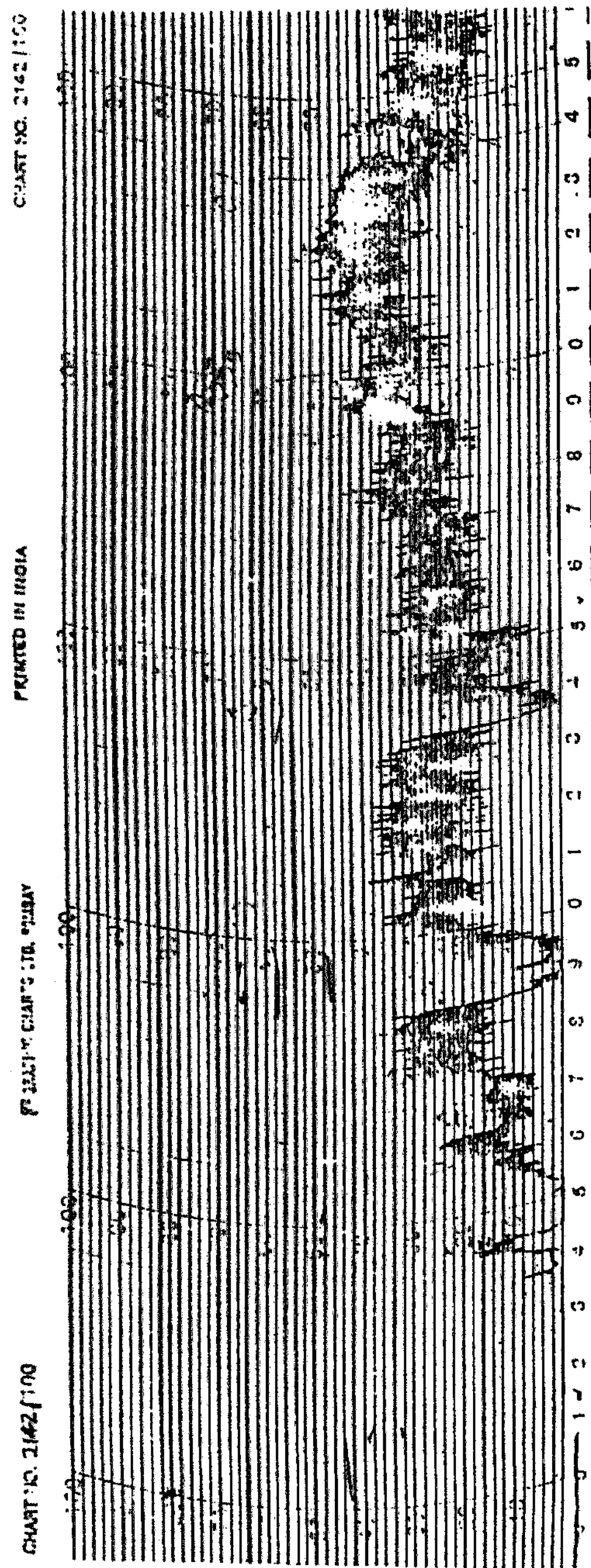


Fig 5a : Typical trace of wind speed showing large variation over short time interval (7-8 July 1992).

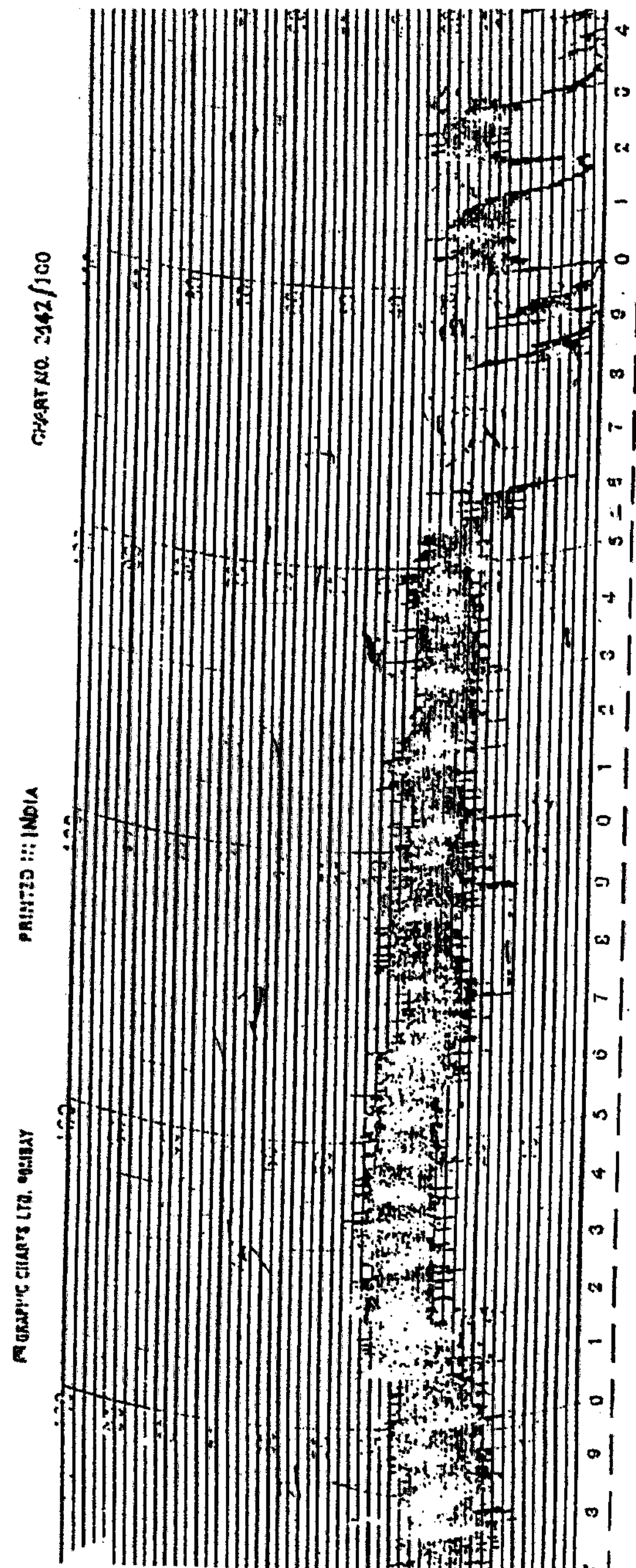


Fig 5b : Typical trace of wind speed showing large variation over short time interval (7-8 July 1992).

Table IV: History of Each Blizzard

Month	S.No	Commencement		Cessation		Dura- tion (hr)	Pressure (hpa)				Pressure drop (hPa)	Temperature (°C)				Rise in temp.	Max. wind (knots) Steady/ gust.
		Date	Time (UTC)	Date	Time (UTC)		Max	Date/ Time	Min.	Dale/ Time		Max.	Date Time	Min.	Date/ Time		
Jan	Nil																
Feb	Nil																
Mar	1	02	1030	03	1800	31.5	995.6	01/06	971.9	02/18	23.7	+1.0	-	-5.1	-	6.1	50/70
	2	05	0300	08	1000	79	995.0	05/09	974.1	08/00	20.9	-3.6	-	-8.2	-	4.6	30/40
	3	10	0300	11	0600	27	987.1	09/21	978.5	11/21	08.6	-3.6	06/10	-9.7	03/11	6.1	36/45
	4	15	1800	16	1100	17	988.2	15/00	980.6	16/09	7.6	-4.4	16/12	-7.8	16/00	3.4	32/46
	5	28	1100	30	1000	47	992.6	27/21	966.6	30/06	26.0	-1.4	30/11	-11.2	27/22	9.8	50/78
Apr	1	06	1030	07	0330	17	995.8	06/00	979.6	08/09	16.2	-8.0	07/12	-13.2	06/12	5.2	30/42
	2	09	0500	09	1230	75	990.8	09/12	979.6	08/09	11.2	-3.6	09/12	-9.8	08/18	6.2	26/34
	3	14	1600	18	0900	89	978.0	14/09	952.2	17/00	25.8	-3.2	16/11	-12.5	14/0730	9.3	45/60
	4	22	0850	25	1030	73.5	990.2	22/09	970.8	24/18	19.4	-6.2	23/1330	-15.0	22/12	8.8	40/60
May	1	01	0200	03	1700	63	1001.1	01/00	996.0	02/03	5.1	-6.0	03/09	-11.5	01/03	5.0	32/38
	2	14	0730	17	1630	81	1006.6	14/00	979.4	15/21	27.2	-6.2	15/18	-14.6	14/0130	8.4	56/71
Jun	1	09	0900	13	1600	103	991.5	08/21	968.2	11/03	23.3	-4.0	10/22	-11.4	08/22	7.4	50/73

Contd.

Table IV : History of Each Blizzard ----*Contd.*

Month	S.No	Commencement		Cessation		Dura- tion (hr)	Pressure (hpa)				Pressure drop (hPa)	Temperature(°C)				Rise in temp.	Max. wind (knots) Steady/ gust
		Date	Time (UTC)	Date	Time (UTC)		Max.	Date/ Time	Min.	Date/ Time		Max	Date/ Time	Min.	Date/ Time		
Jul	1	12	0300	14	0300	48	996.2	11/03	987.9	12/06	4.3	-9.7	13/10	-13.0	12/2230	3.3	36/48
	2	25	2200	26	2200	24	976.8	23/18	955.8	25/03	21.0	-16.0	26/00	-28.0	25/03	12.0	32/45
	3	27	1800	31	1230	90.5	978.9	27/15	961.4	31/06	17.5	-15.5	30	-18.0	29/22	2.5	28/41
Aug	1	07	0200	08	0130	23.5	976.9	06/03	968.0	07/00	8.9	-6.8	08/01	-11.8	06/12	5.0	30/44
	1	09	1630	12	0330	59	999.5	09/12	955.1	11/15	44.4	-10.6	12/01	-16.3	09/12	5.7	56/80
Sep	1	31 Aug	2130	03	1630	67	997.4	03/21	967.1	29 Aug/15	30.3	-9.0	03/1330	-23.7	31 Aug/18	14.7	38/50
	2	04	1700	05	1300	20	997.4	03/21	988.9	05/00	8.5	-10.6	04	-14.8	05/00	4.2	37/54
Oct	1	09	0115	09	0930	8.5	978.3	08/09	964.3	08/21	14.0	Variation is due to diurnal cycle				-	32/40
	1	16	1400	18	1000	44	1003.0	16/03	978.3	17/18	24.7	-3.0	18/11	-16.4	16	13.4	44/64
	3	22	0200	22	1230	10.5	993.5	21/06	981.8	22/12	11.7	Variation is due to diurnal cycle				-	28/42
	4	24	1230	26	0700	42.5	999.8	24/00	974.5	25/21	25.5	-3.5	24/16	-9.0	24/0630	5.5	42/58
Nov	1	07	2200	09	1600	42	984.5	07/12	959.0	09/09	25.5	Variation due to diurnal cycle				-	46/62
Dec	1	13	2000	14	2000	24	985.9	12/18	970.1	14/03	15.8	-2.6	13/-	-6.6	13/-	4.0	42/60

depend on it. In this report, for defining blizzard, three criteria are used. In order of significance of relative weightage, these are:

- i) Wind speed >30 knots
- ii) Present weather, WW, either 38 or 39 (blowing snow)
- iii) Visibility, VV, <94 (1000 m)

An ideal blizzard meets all the three conditions simultaneously. However, quite alien, it may not be so; either a particular criterion is not satisfied or the conditions may not be in the appropriate range.

Maximum and minimum values of pressure and temperature have been picked up from the entire sequence of weather leading to, and during the blizzard to get a picture of the magnitude of pressure drop and rise in temperature - the characteristic patterns of change in association with blizzards. During blizzards, air flow is extremely turbulent, gustiness increasing with wind speed. Hence, in this report both mean wind speed and gust are given (Table IV).

History of each blizzard experienced during 1992 is given in Table IV. Blizzards are less frequent during summer months (Nov., Dec., Jan.) and early autumn (Feb.). But, as autumn advances, the frequency of blizzards increases. During winter (April to August) blizzards occur at an average rate of two per month. The frequency increases further in late spring (October).

Maximum wind during blizzards ranged from 35 to 80 knots. The drop in pressure related to blizzards ranged from 7.6 hPa to 44.4 hPa with an average value of about 20 hPa. The rise in temperature ranged from 2.6°C to 17.5°C, average being 8.0°C. The shortest blizzard was of 7.5 hour duration while the longest one lasted for 103 hours. On an average blizzards last for 24 to 48 hours. It is found that blizzards at Maitri which is ~100 kms from the shelf boundary, are less frequent, less severe and short lived compared to the same parameters as experienced at Dakshin Gangotri station which is only 16 kms from the shelf boundary.

Commonly, blizzards are associated with bad weather spells - overcast sky with or without snowfall and easterly/southeasterly wind. However, on certain rare occasions, blizzards result from strong southerly wind accompanied by clear sky. At Dakshin Gangotri, southerly wind is much less frequent and its speed is never strong enough to give rise to blizzard.

e) Case study of a blizzard which prevailed from 2200 UTC of 25th July 1992 to 2200 UTC of 26th July 1992.

The blizzard under study has an uncommon feature of high southerly wind changing over to high southeasterly wind (backing) with a short transition of

calm. It also highlights the role of wind direction in producing changes in weather conditions. The changes in meteorological parameters in association with this blizzard are shown in Fig 6. Pressure started falling from 00 UTC of 24th and dropped by 17 hPa in 24 hours. Moderately high southerly wind prevailed during the whole day on 24th. The sky was clear and temperature did not vary much although it showed slight diurnal tendency at a time when the Sun had returned only a few days back. Visibility was excellent. By 00 UTC of 25th southerly wind died down and calm condition prevailed for several hours. Then southeasterly wind set-in (12 UTC) progressively picking up speed. Change in wind direction resulted in a sharp increase in cloud cover (WW=03). Temperature increased continuously (11°C in 24 hours). By 18 UTC snowfall commenced (WW=73 and VV=96). With wind speed reaching 30 knots by 00 UTC of 26th, loose snow started blowing reducing visibility to 2 m (VV=90). The blizzard had commenced at 22 UTC on the 25th July.

f) Cloud Cover

Cloud cover over Maitri occurs mainly under the influence of depressions moving east-southeastward between latitudes 65° and 70°S. On rare occasions isolated orographic clouds occur due to southerly wind blowing over comparatively shallow mountain range. For a few weeks during peak summer cumulus development apparently of continental origin occurs. Mean daily cloud amount in different months is given in Table I.

Cloud cover under the influence of low pressure systems shows a typical pattern (Fig. 7) of alternating sequence of clear sky changing over to overcast and then clearing as the system moves away. Changes in cloud amount are often rapid especially as the system approaches and moves away (transition periods). Usually high clouds appear first. Typical high clouds are cirrus in bands converging toward one or two opposite points of the horizon either below or extending more than 45° above the horizon. They grow denser and eventually cirrostratus overcast may result. Later, altostratus followed by stratocumulus overcast generally results. Often stratocumulus overcast is so dense and continuous that individual elements cannot be distinguished.

Isolated orographic clouds are in the form of fishes or almonds (altocumulus lenticularis) when the wind is southerly or southeasterly caused by low level waves in the air flow. This results due to obstruction by the Wohlthat mountains at a distance of about 100 kms SSE of Maitri station. Since these winds blow over a vast ice plateau, the air is very dry and the cloud amount varies from just a few specs to 2 octas. Although typical lenticularis clouds are altocumulus, over Maitri the height of these clouds often falls in the low category.

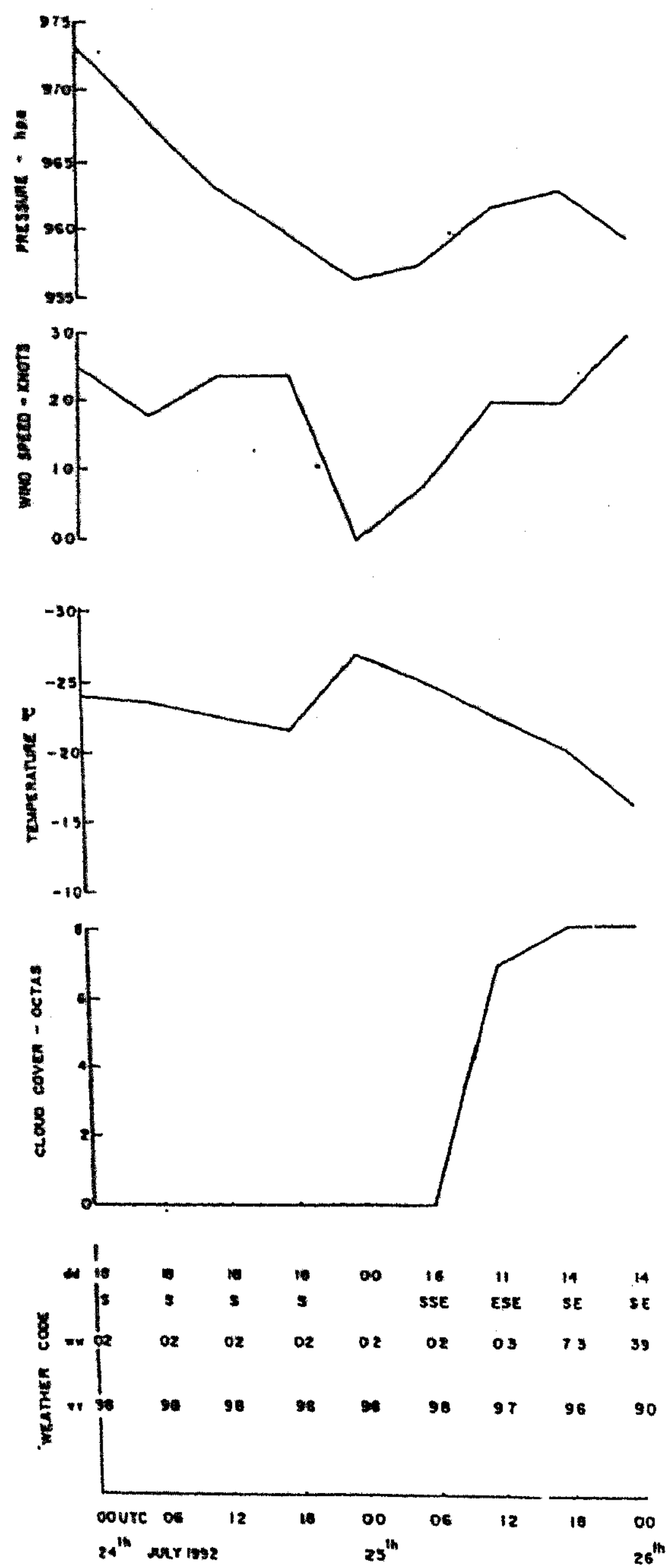


fig 6: Changes in meteorological parameters in association with a blizzard.

JULY 1992

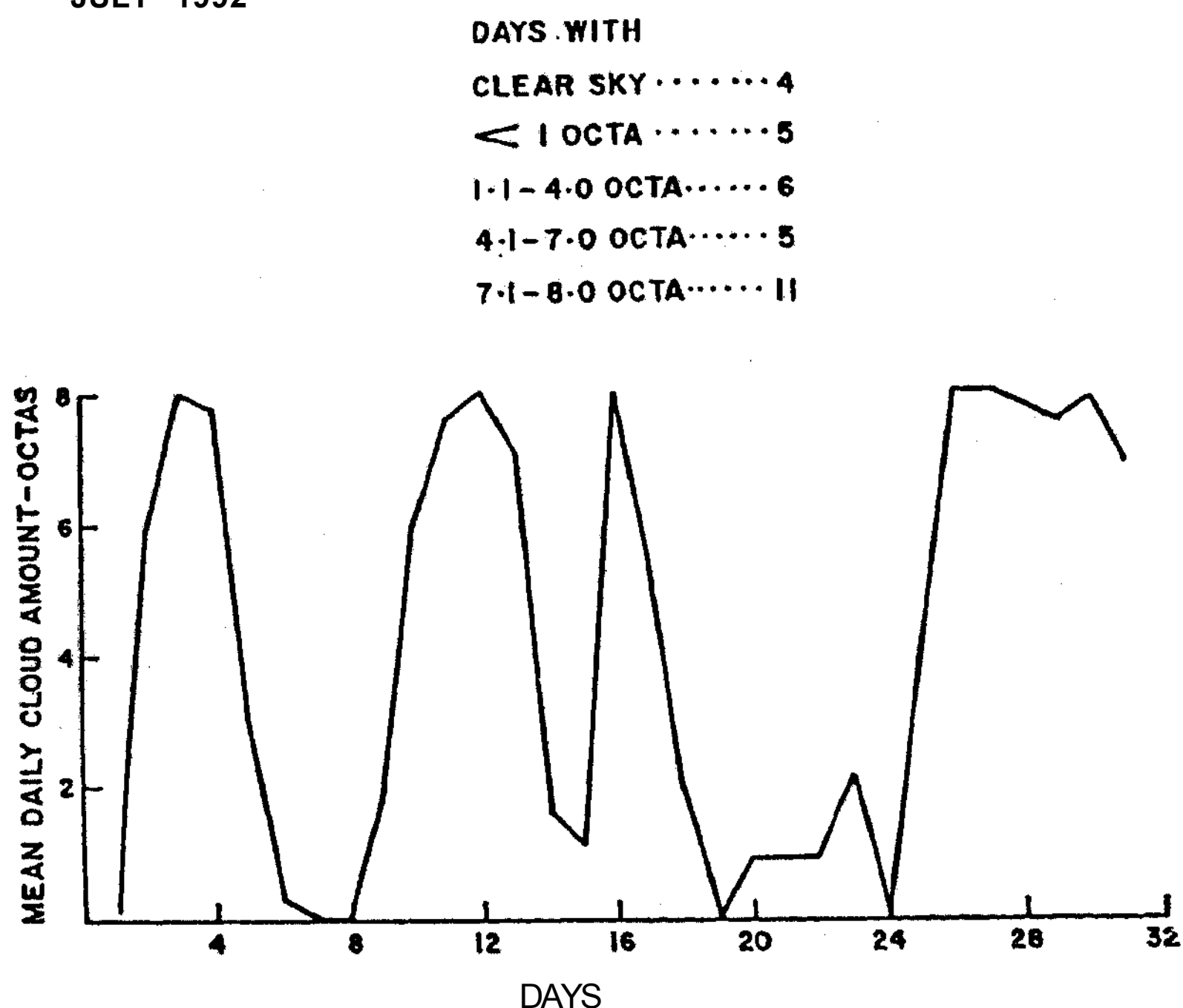


Fig 7: Typical occurrence of cloud cover at Maitri, Antarctica.

Cumulus development is observed on several days from the third week of November (beginning of continuous sunlight period) to third week of December (summer solstice). Invariably cumulus and stratocumulus other than that formed from the spreading out of cumulus ($C_L=8$ in surface weather code), are seen together. However, base of the two layers are at different levels, in the range of 150 to 300 m. These clouds are observed between 0900 and 1800 hrs, maximum development to the extent of 6 octas occurring between 1300 and 1600 hrs.

g) *Global Solar Radiation*

At Maitri, in the year 1992, first sunset occurred on January 26, annual sunset was on May 24; Sun returned on July 19 and remained above horizon from November 22.

Mean hourly global solar radiation values in different months are given in Table V. Mean daily radiation was maximum in December and minimum in

Table V : Mean Hourly Global Solar Radiation (Mega Joule m⁻²) in Different Months (1992)

Hours	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
1	*	-	-	-	-	-	-	-	-	-	0.04	0.14
2	0.07	0.03	-	-	-	-	-	-	-	-	0.07	0.18
3	0.09	0.04	-	-	-	-	-	-	-	0.02	0.14	0.32
4	0.32	0.11	0.01	-	-	-	-	-	-	0.12	0.38	0.51
5	0.23	0.23	0.07	-	-	-	-	-	0.01	0.16	0.70	0.81
6	0.34	0.47	0.08	-	-	-	-	-	0.05	0.35	1.03	1.04
7	0.57	0.78	0.22	0.03	-	-	-	0.03	0.30	0.41	1.45	1.42
8	0.80	1.06	0.39	0.15	-	-	-	0.11	0.52	0.98	1.77	1.69
9	0.94	1.41	0.55	0.16	-	-	-	0.24	0.77	1.32	2.11	2.17
10	1.14	1.66	0.76	0.30	0.03	-	-	0.28	0.94	1.54	2.38	2.41
11	1.49	1.87	0.90	0.43	0.07	-	*	0.31	1.10	1.73	2.41	2.52
12	1.72	1.90	0.94	0.48	0.08	-	*	0.30	0.75	1.87	2.59	2.63
13	1.80	1.79	1.00	0.42	0.09	-	*	0.21	0.34	1.80	2.57	2.51

Contd.

Table V : Mean Hourly Global Solar Radiation (Mega Joule m ²) in Different Months (1992) - <i>Contd.</i>												
Hours	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec
14	1.70	1.56	0.88	0.36	0.05	-	*	0.13	0.77	1.63	2.36	2.29
15	1.48	1.46	0.74	0.25	0.02	-	*	0.09	0.53	1.38	2.20	2.06
16	1.46	1.20	0.56	0.19	-	-	*	0.01	0.33	1.12	1.83	1.66
17	1.23	0.95	0.35	0.18	-	-	-	-	0.23	0.80	1.49	1.42
18	0.79	0.70	0.16	-	-	-	-	-	0.16	0.65	1.16	1.06
19	0.50	0.38	0.07	-	-	-	-	-	0.09	0.33	0.90	0.82
20	0.37	0.20	0.05	-	-	-	-	-	-	0.22	0.60	0.60
21	0.20	0.11	-	-	-	-	-	-	-	0.06	0.33	0.40
22	0.10	0.04	-	-	-	-	-	-	-	-	0.22	0.28
23	*	-	-	-	-	-	-	-	-	-	0.17	0.17
24	*	-	-	-	-	-	-	-	-	-	0.07	0.13
Total	17.34	17.91	7.73	2.95	0.34	-	-	1.71	6.89	16.49	28.97	29.24

* No data. All other blanks indicate no sunshine.

May. There is no data for July due to continuous bad weather after return of Sun.

An attempt was made to record the diffused radiation by shading the pyranometer at Fixed intervals with the help of a shading device on a clear sky day (March 17) close to the autumnal equinox. The idea was to segregate direct and diffused components of the total radiation. The data are given in Table VI and shown graphically in Fig 8. It was found that the two components account for 66 and 34 per cent respectively of the total radiation.

h) Vertical Distribution of Ozone

During 1992, a total of 45 ozone ascents were taken, more or less at weekly intervals. Important features of stratospheric ozone are given In Tables VII and VIII. Ozone profiles for 31.8.92 and 18.10.92 have been shown in Fig 9(a). Similarly those for 18.10.92 and 24.11.92 are depicted in Fig 9(b). The two figures bring out the depletion and subsequent filling up of stratospheric ozone over Maitri in less than three months during spring and early summer.

**Table VI: Direct and Diffuse Components of Global Solar Radiation in tie Vicinity of Autumnal Equinox at Maitri, Antarctica
(Date : 17 March 1992; Unit : Mega-Joule m⁻²)**

Hour	Total	Diffuse	Direct
6-7	0.26	0.13	0.13
7-8	0.62	0.21	0.41
8-9	0.90	0.31	0.59
9-10	1.19	0.39	0.80
10-11	1.35	0.44	0.91
11-12	1.43	0.47	0.96
12-13	1.45	0.42	1.03
13-14	1.40	0.40	1.00
14-15	1.23	0.40	0.83
15-16	0.87	0.35	0.52
16-17	0.54	0.26	0.28
17-18	0.22	0.14	0.08
Total	11.46	3.92(34.2%)	7.54(65.8%)

Table VII: Ozone Partial Pressure in mb at Selected Altitudes in Different Months											
Pressure level (hPa)	22.1.92	5.2.92	2.4.92	26.5.92	15.6.92	31.8.92	18.10.92	2.11.92	24.11.92	7.12.92	23.12.92
300	11	38	33	8	17	22	14	19	12	10	32
200	19	48	46	18	54	48	34	44	25	32	29
100	64	91	107	56	130	128	9	14	20	28	92
70	110	119	119	133	145	156	8	23	79	119	104
50	125	127	121	149	145	163	56	53	100	148	123
30	107	106	85	110	20	65	5	89	151	125	130

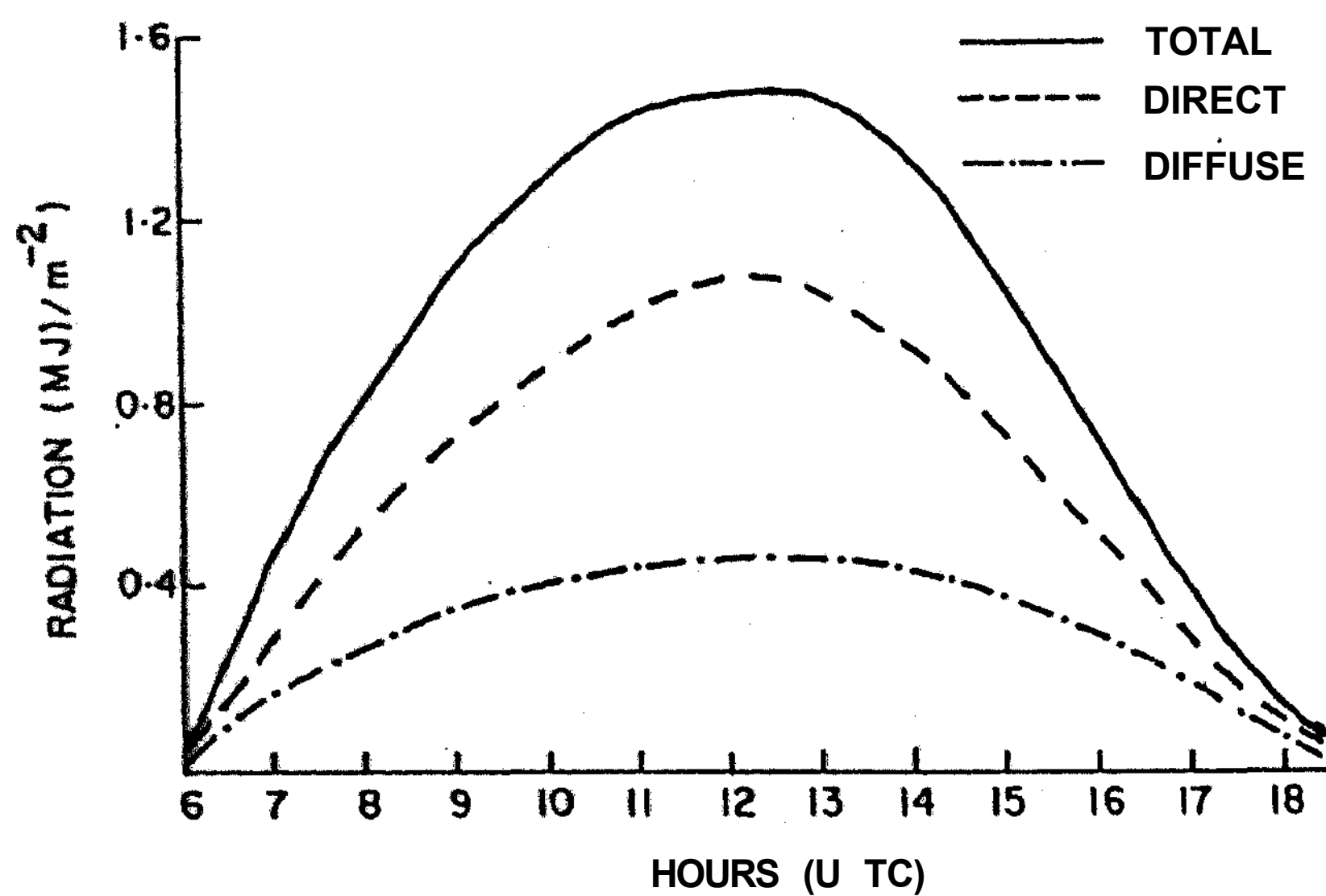


Fig 8 : Total direct and diffused radiation on March 17,1992 at Maitri station, Antarctica.

Table VHI:Ozone Maximum Level and Ozone Maximum Concentration in Different Months

Month	Ozone Max Level (hPa)	Ozone Max. Pressure (mb)
Jan (22nd)	46	134
Feb (5th)	48	128
Mar (1 st)	63	128
Apr (2nd)	91	127
May (26th)	50	149
Jun (15th)	79	149
Jul (18th)	59	126
Aug (31st)	52	167
Sep (30th)	24	75
Oct (18th)	40	93
Nov (2nd)	28	90
Nov (24th)	26	154
Dec (7th)	45	149

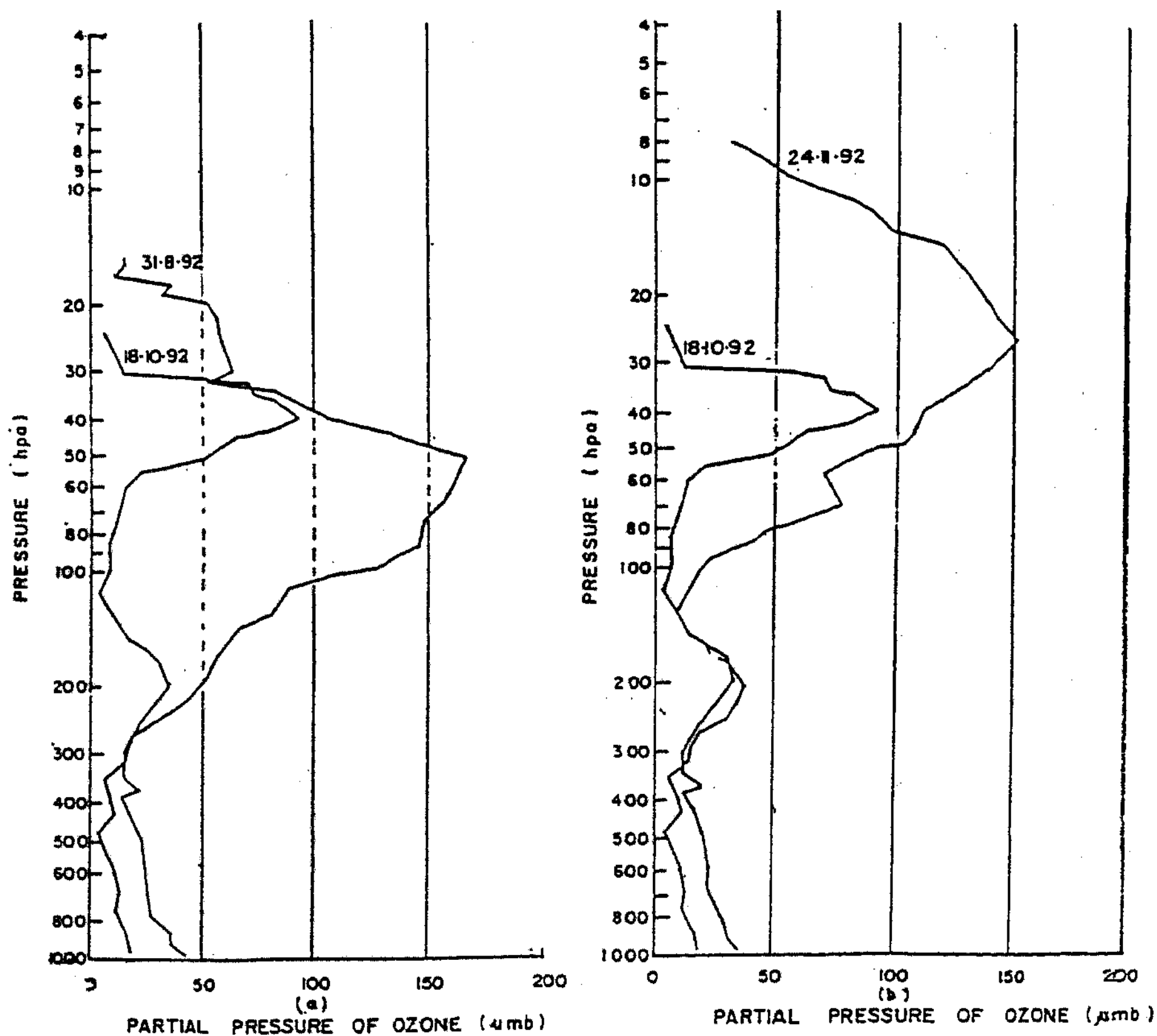


Fig 9: Ozone profiles over Maitri, Antarctica - a) before depletion and b) after recovery in comparison with depleted one.

The temperature profiles for these three dates are shown In Fig 10. From the results it is seen that the phenomenon of stratospheric ozone depletion is strongly related to stratosphere temperature. It appears that colder the winter stratosphere, greater is the depletion of ozone in the ensuing spring.

Ozone concentration does not vary much In the troposphere (upto 8-9 kms) and lower stratosphere (9-16 kms). The small variations from month to month do not show any seasonal trend. It is in the middle stratosphere (16-24 kms) that maximum depletion of ozone occurs in spring. While ozone maximum level varies between 90 and 50 hPa (18-21 kms) in all other seasons, it is about 30 hPa (-24 kms) in the beginning of summer (November) indicating that ozone build-up starts from top and proceeds downwards as summer advances.

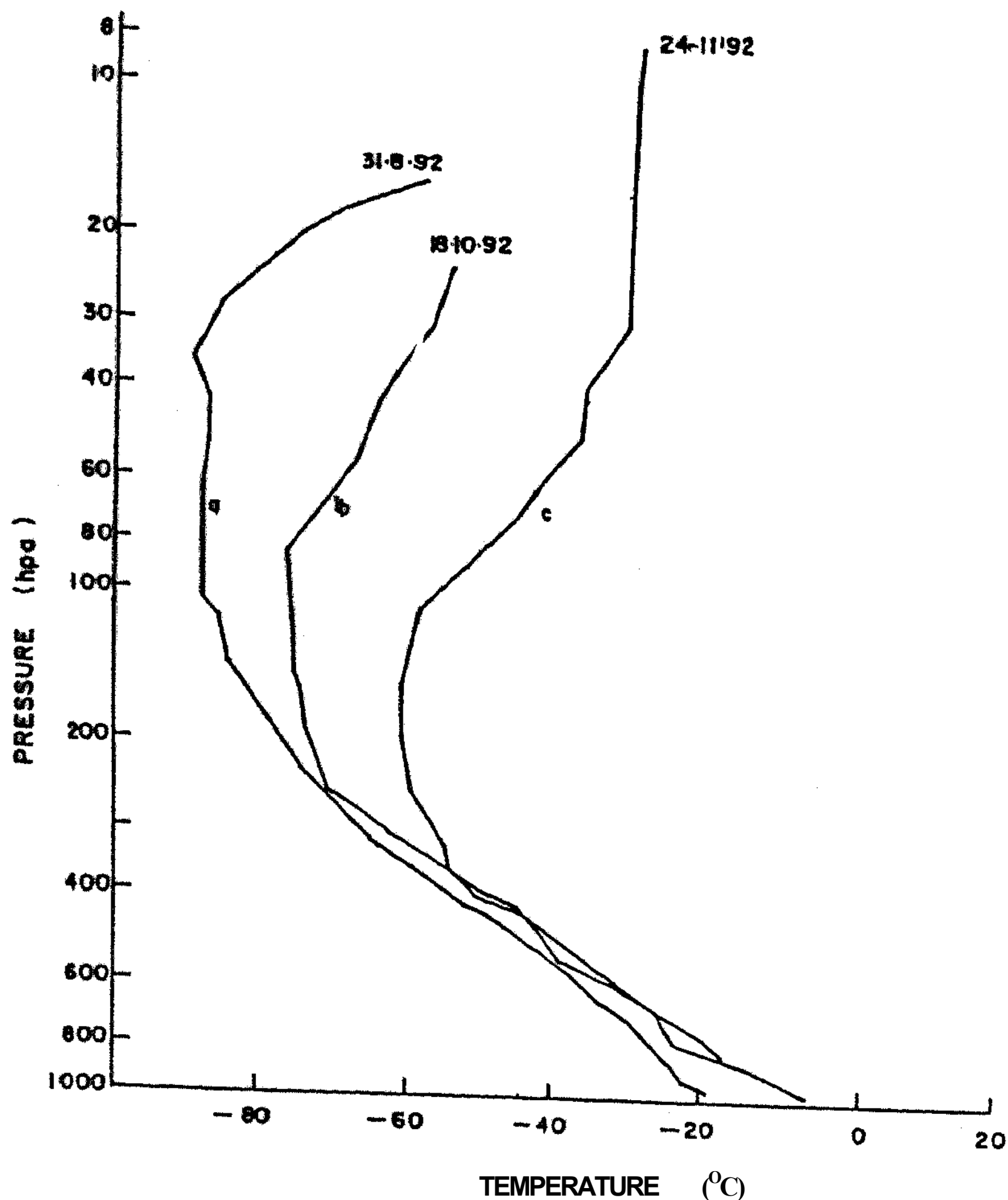


Fig 10: Temperature profiles—a) at the end of winter, b) during spring and c) beginning of summer.

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